Project Proposal

Why

As I'm set to do my co-op at NetSkrt in January, I wanted to learn a bit more about their systems and business problem. So I planning to do a small scale edge CDN with caching proxy for HLS video streaming, and for the 7171 component, a Star Schema data warehouse that handles delivery logs for analytics to optimize cache efficiency for the 8071 component.

What

I'm wanting to do a project on CDNs, with a focus in video streaming. The goal of the project is simulate a edge CDN. The plan for this is to develop a Edge Cache Node, that uploads data to the analytics warehouse, and test it with some simulators. Everything that would be a hardware component is going to be represented as a docker container, to reenact the bare metal that would be in place. I likely won't create enough data from testing to populate the data warehouse adequately so I likely seed it from some other source.

Major Deliverables

- Origin Server -> to act as the sender/source for the content that's contained on the edge cache
- Edge Cache Node -> where the actual caching takes place and the source of the data for the data warehouse
- Receiver Server -> likely just some scripting, like Jmeter to act as a bunch of users
- Data warehouse

1. What Data Ware House you are creating?

> A CDN Video Cache and Streaming Analytics warehouse

2. Why? list the associations/correlations/classifications/trends you want to learn.

> Correlations/Associations
>
> - Correlations between playback time and viewer drop-off rate
> - Correlations between when a cache should expire and traffic against
> - Associations between demand and genre

> Classifications
>
> - Content and quality
> - Temporal Windows

> Trends
>
> - Demand during peak hours

3. Capture the above needs in the form of OLAP queries (in simple English). Just couple of queries would be sufficient.

> What is the total bandwidth save for each cache hit of each video how many unique viewers reached each successive chunk segment, and at what segment index does the sharpest viewer drop-off

> happen

4. OLAP Schema(s) of your data warehouse

> fact_chunk_delivery
>
> • delivery_id • time_id • content_id • client_id • eviction_policy • is_cache_hit • bytes_used

> • dim_time: time_id , timestamp, hour, minute • dim_content: content_id, video_id, video_title, genre, duration_sec, bitrate • dim_client: client_id, session_id

5. SQL representation of your OLAP queries

```sql
select
  dc.video_title,
  dt.hour,
  count(*)
  sum(f.is_cache_hit)
  round(100.0 * sum(f.is_cache_hit) / count(*), 1)
from fact_chunk_delivery f
join dim_content dc on f.content_key = dc.content_key
join dim_time dt on f.time_key = dt.time_key
group by rollup (dc.video_title, dt.hour)
order by dc.video_title nulls last, dt.hour nulls last;
```

6. Data source(s) for your data warehouse

> Kaggle, youtube, etc, if I need to, otherwise `ffmpeg` and tracking setup from the CDN

7. Technology to be used used in the creation of your data engineering solution including for ETL, backend server and database, and plotting/reporting

> Python/Pyspark For ETL, backend in rust.